Chapter 12 - SoundChip and SFX

The SoundChip is Intuition Engine's primary synth. It has ten flexible channels - each one a full oscillator with envelope, sweep, ring-modulation, hard-sync, PWM, and DAC inputs - plus four SFX sample channels. All channels are mixed into the global mixer described in Chapter 11.

This chapter describes one channel in detail, then lists the SFX channel registers, then shows the BASIC `SOUND` keyword's options.

12.1 A SoundChip channel

A channel is a `0x40`-byte block. Inside one channel:

Offset	Name	Bits	Purpose
`0x00`	`FREQ`	32	Frequency in 16.8 fixed-point Hz.
`0x04`	`VOL`	8	Output volume, `0`–`255`.
`0x08`	`CTRL`	8	Gate and per-channel control. Bit `1` = gate.
`0x0C`	`DUTY`	8	PWM duty (square wave) or pulse parameter.
`0x10`	`SWEEP`	8	Sweep enable / direction / period / shift.
`0x14`	`ATK`	8	Envelope attack.
`0x18`	`DEC`	8	Envelope decay.
`0x1C`	`SUS`	8	Envelope sustain.
`0x20`	`REL`	8	Envelope release.
`0x24`	`WAVE_TYPE`	8	Waveform select (see below).
`0x28`	`PWM_CTRL`	8	Bit `7` = PWM enable, bits `0`–`6` = rate.
`0x2C`	`NOISEMODE`	8	Noise generator algorithm (see below).
`0x30`	`PHASE`	8	Write to reset the oscillator phase.
`0x34`	`RINGMOD`	8	Bit `7` = enable, bits `0`–`2` = source channel.
`0x38`	`SYNC`	8	Bit `7` = enable, bits `0`–`2` = source channel.
`0x3C`	`DAC`	8	Signed 8-bit sample (bypasses oscillator + envelope).

12.1.1 Channel locations

Ten channels are mapped in three blocks:

Channels	Base	Stride	End
`0`–`3`	`0xF0A80`	`0x40`	`0xF0B7F`
`4`–`6`	`0xF0C40`	`0x40`	`0xF0CFF`
`7`–`9`	`0xF0D40`	`0x40`	`0xF0DFF`

Channels `0`–`3` are the **primary** group used by BASIC's `SOUND` keyword. Channels `4`–`6` and `7`–`9` are the "SID2" and "SID3" voice groups (Chapter 15) that give the chip three SID voices on top of its primary four. Every channel uses the same register layout shown in §12.1.

12.1.2 Waveforms

Each channel's `WAVE_TYPE` register selects one of five waveforms. The values are zero-based:

WAVE_TYPE	Waveform
0	Square (with `DUTY` controlling pulse width)
1	Triangle
2	Sine
3	Sawtooth
4	Noise (algorithm chosen by `NOISEMODE`)

12.1.3 Noise modes

When `WAVE_TYPE = 4`, the noise generator is configured by `NOISEMODE`:

NOISEMODE	Algorithm
0	White (LFSR, default)
1	Periodic / loop
2	Metallic
3	PSG-style (AY/YM)
4	TED 8-bit
5	SN76489 15-bit white
6	SN76489 15-bit periodic
7	SN76489 16-bit white
8	SN76489 16-bit periodic

This is how the SoundChip reproduces the noise characteristics of the other chips' channels when used inside a single track.

12.1.4 Envelopes

Each channel has its own ADSR envelope, parameterised by `ATK`, `DEC`, `SUS`, `REL`. Five envelope shapes are selectable through the shared `ENV_SHAPE` register at `0xF0804`:

ENV_SHAPE	Behaviour
0	Standard ADSR (default).
1	Saw-up: linear rise to `1.0`, then hold.
2	Saw-down: linear fall to `0.0`, then hold.
3	Loop: ADSR but loops after release.

ENV_SHAPE	Behaviour
4	SID-style exponential ADSR.

Per-channel shape registers live at `ENV_SHAPE_CH_BASE + ch*4` (`0xF0860`).

12.1.5 Sweep, ring-mod, sync

The `SWEEP` register encodes a pitch sweep:

Bit	Field	Meaning
7	Sweep enable	`0` = off, `1` = on.
3	Direction	`0` = up, `1` = down.
4–6	Period	Speed (`0`–`7`).
0–2	Shift	Amount per step (`0`–`7`).

`RINGMOD` selects another channel as the ring-modulator source; when bit `7` is set, the channel's output is multiplied by the source's instantaneous amplitude. `SYNC` does the same for hard sync: when the source crosses zero, the channel's oscillator is forced to reset its phase.

12.1.6 DAC mode

Writing a signed 8-bit value to a channel's `DAC` register overrides the oscillator and envelope for that sample and outputs the byte directly. This lets a CPU drive the channel as a raw sample player.

12.2 The SFX channels

In addition to the ten synth channels, the SoundChip has four **SFX channels** that play samples from main memory. They live in their own register block:

Address	Channel	Stride
`0xF0E80`	`0`	`0x20`
`0xF0EA0`	`1`	`0x20`
`0xF0EC0`	`2`	`0x20`
`0xF0EE0`	`3`	`0x20`

Inside one channel:

Offset	Name	Purpose
`0x00`	`SFX_PTR`	Sample data address.
`0x04`	`SFX_LEN`	Sample length, in bytes.
`0x08`	`SFX_LOOP_PTR`	Loop start (if looped).
`0x0C`	`SFX_LOOP_LEN`	Loop length.
`0x10`	`SFX_FREQ`	Playback rate in Hz.
`0x14`	`SFX_VOL`	Volume, `0`–`255`.

Offset	Name	Purpose
0x18	SFX_FORMAT	0 = signed 8-bit, 1 = unsigned 8-bit, 2 = signed 16-bit.
0x1C	SFX_CTRL	Bit 0 = trigger, bit 1 = stop, bit 2 = loop enable.

The block ends at `0xF0EFF`.

`SFX_CTRL` is also the status read-back: bit `0` = playing, bit `1` = error.

12.3 BASIC keywords

The `SOUND` keyword groups several subcommands. See Chapter 2 for full syntax.

Form	Effect
`SOUND`*ch*`,`*freq*`,`*vol*`[ ,`*wave*`[ ,`*duty*`]]`	Set frequency, volume, and (optionally) waveform and duty cycle on channel `0`–`3`.
`SOUND FILTER`*cutoff*`,`*resonance*`,`*type*	Configure the global filter.
`SOUND REVERB`*mix*`,`*decay*	Configure the global reverb.
`SOUND OVERDRIVE`*amount*	Configure the global overdrive.
`SOUND NOISE`*ch*`,`*mode*	Set the noise mode on channel `0`–`3`.
`SOUND WAVE`*ch*`,`*type*	Set the waveform type on channel `0`–`3`.
`SOUND SWEEP`*ch*`,`*enable*`,`*period*`,`*shift*	Configure a pitch sweep.
`SOUND SYNC`*ch*`,`*source*	Configure hard sync.
`SOUND RINGMOD`*ch*`,`*source*	Configure ring modulation.
`SOUND PLAY "`*file*`"[,`*subsong*`]`	Play a music file (Chapter 22).
`SOUND STOP`	Stop the current music.
`ENVELOPE`*ch*`,`*atk*`,`*dec*`,`*sus*`,`*rel*	Set the ADSR envelope on channel `0`–`3`.
`GATE`*ch*`, ON` / `GATE`*ch*`, OFF`	Open or close a channel's gate.

`SOUND` writes 16.8 fixed-point frequency (`Hz × 256`) into the `FREQ` register and the volume into `VOL`. The optional waveform and duty arguments go into `WAVE_TYPE` and `DUTY`.

12.4 A short example

A BASIC fragment that plays a sustained square-wave A4 (440 Hz) through channel `0`:

```
10 POKE &H000F0800, 1                 : REM AUDIO_CTRL = on
20 ENVELOPE 0, 50, 100, 200, 100      : REM ADSR
30 SOUND 0, 440, 200, 0, 128          : REM square, duty = 128
40 GATE 0, ON
```

To stop the note cleanly, `GATE 0, OFF`. To stop it abruptly, write `0` to `SOUND 0, 0, 0`.

12.5 Putting it together

A SoundChip channel is the same kind of object whichever number it has. Most programs use channels `0–3` for melody, then layer percussion or noise on the higher channels. The SFX channels are the fastest way to play short samples (gunshots, beeps, voice) without disturbing the synth voices.

The next chapter covers the AY-3-8910 / YM-2149 PSG, the first of the heritage engines.